Applied Statistical Analysis I

Multiple linear regression

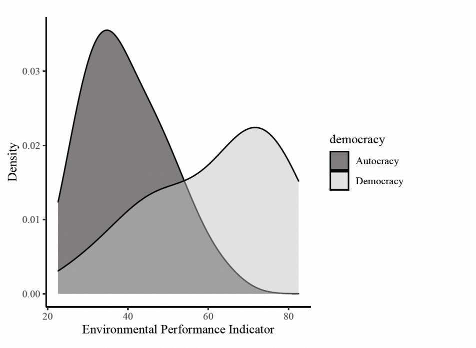
Elena Karagianni, PhD Candidate
karagiae@tcd.ie

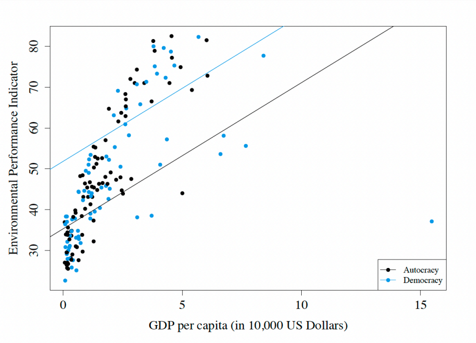
November 26, 2025

Department of Political Science, Trinity College Dublin

Categorical Independent Variables

Categorical Independent Variables





Categorical IVs: Multiple Categories

$$\text{Environmental Performance}_i = \alpha + \beta_1 \times \text{Income} + \beta_2 \times \text{Region}_i + \epsilon_i$$

```
## table(qog_data$ht_region)
##
##                   Eastern Europe (1)               Latin America(2)
##                                   28                             20
##    North Africa & the Middle East (3)        Sub-Saharan Africa (4)
##                                   20                             49
##  Western Europe and North America (5)                East Asia (6)
##                                   27                              6
##                  South-East Asia (7)              South Asia (8)
##                                   11                              8
##                      The Pacific (9)            The Caribbean (10)
##                                   12                             13
```

Categorical IVs: in R

```r
# Load package
library(fastDummies)

# Create dummy variables for categorical variable
qog_data <- dummy_cols(qog_data,
                       select_columns = c("ht_region"))

# Print first 5 rows in dataset
head(qog_data[c("ht_region_1",
            "ht_region_2",
            "ht_region_3",
            "ht_region_4",
            "ht_region_5",
            "ht_region_6",
            "ht_region_7",
            "ht_region_8",
            "ht_region_9",
            "ht_region_10")], 5)
```

```
##   ht_region_1 ht_region_2 ht_region_3 ht_region_4 ht_region_5
## 1           0           0           0           0           0
## 2           1           0           0           0           0
## 3           0           0           1           0           0
## 4           0           0           0           0           1
## 5           0           0           0           1           0
##   ht_region_6 ht_region_7 ht_region_8 ht_region_9 ht_region_10
## 1           0           0           1           0            0
## 2           0           0           0           0            0
## 3           0           0           0           0            0
## 4           0           0           0           0            0
## 5           0           0           0           0            0
```

Categorical IVs: in R

```r
# Run regression model
m2 <- lm(epi_epi ~ income +
            ht_region_1 + ht_region_2 + ht_region_3 +
            # no region 4 (Sub-Saharan Africa) = reference category.
            ht_region_5 + ht_region_6 + ht_region_7 + ht_region_8 +  ht_region_9 +
            ht_region_10, data = qog_data)

# Print results
summary(m2)
```

```
Coefficients: (1 not defined because of singularities)
             Estimate Std. Error t value Pr(>|t|)
(Intercept)   32.3992     1.1296  28.683  < 2e-16 ***
income         1.7410     0.4061   4.287 3.23e-05 ***
ht_region_1   18.4245     1.8769   9.817  < 2e-16 ***
ht_region_2   11.6208     2.0362   5.707 6.01e-08 ***
ht_region_3    9.4434     2.4665   3.829 0.000189 ***
ht_region_5   35.2532     2.4854  14.184  < 2e-16 ***
ht_region_6   16.2287     3.6737   4.418 1.91e-05 ***
ht_region_7    4.1247     2.7820   1.483 0.140281
ht_region_8   -2.1694     3.2676  -0.664 0.507774
ht_region_9        NA         NA      NA       NA
ht_region_10  11.0665     3.5607   3.108 0.002257 **
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 7.528 on 149 degrees of freedom
  (35 observations deleted due to missingness)
Multiple R-squared:  0.7897,Adjusted R-squared:  0.777
F-statistic: 62.16 on 9 and 149 DF,  p-value: < 2.2e-16
```

Interpretation in reference to baseline categories

Categorical IVs: in R

```r
# Use relevel to code dummy variables on the fly
# specify region 4 (Sub-Saharan Africa) = reference category
m3 <- lm(epi_epi ~ income + relevel(as.factor(ht_region), ref = "4"),
            data = qog_data)

# Print results
summary(m3)
```

```
                                          Estimate Std. Error t value Pr(>|t|)
(Intercept)                                32.3992     1.1296  28.683  < 2e-16 ***
income                                      1.7410     0.4061   4.287 3.23e-05 ***
relevel(as.factor(ht_region), ref = "4")1  18.4245     1.8769   9.817  < 2e-16 ***
relevel(as.factor(ht_region), ref = "4")2  11.6208     2.0362   5.707 6.01e-08 ***
relevel(as.factor(ht_region), ref = "4")3   9.4434     2.4665   3.829 0.000189 ***
relevel(as.factor(ht_region), ref = "4")5  35.2532     2.4854  14.184  < 2e-16 ***
relevel(as.factor(ht_region), ref = "4")6  16.2287     3.6737   4.418 1.91e-05 ***
relevel(as.factor(ht_region), ref = "4")7   4.1247     2.7820   1.483 0.140281
relevel(as.factor(ht_region), ref = "4")8  -2.1694     3.2676  -0.664 0.507774
relevel(as.factor(ht_region), ref = "4")10 11.0665     3.5607   3.108 0.002257 **
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 7.528 on 149 degrees of freedom
  (35 observations deleted due to missingness)
Multiple R-squared:  0.7897,Adjusted R-squared:  0.777
F-statistic: 62.16 on 9 and 149 DF,  p-value: < 2.2e-16
```

Under control of income, Eastern Europe has an Environmental Performance Index score of 18.4245 scale points higher than Sub-Saharan Africa.

Using Dummy Variables for Multiple Categories

- Dummy variable trap:
    - Base group is represented by the intercept.
    - Adding all dummies $\rightarrow$ perfect multicollinearity.

Using Dummy Variables for Multiple Categories

- Dummy variable trap:
    - Base group is represented by the intercept.
    - Adding all dummies $\rightarrow$ perfect multicollinearity.
- Solution: Use $k - 1$ dummies
- Interpretation always **relative to baseline**.

Using Dummy Variables for Multiple Categories

- Dummy variable trap:
  - Base group is represented by the intercept.
  - Adding all dummies $\rightarrow$ perfect multicollinearity.
- Solution: Use $k-1$ dummies
- Interpretation always **relative to baseline**.

	Dummy Variables		
Social Class	D_1	D_2	
lower	0	0	$\hat{Y} = \hat{\beta}_0$
middle	1	0	$\hat{Y} = \hat{\beta}_0 + \hat{\beta}_1$
upper	0	1	$\hat{Y} = \hat{\beta}_0 + \hat{\beta}_2$

Using Dummy Variables for Multiple Categories

- What if we want to test the difference between middle and upper class?
- Cleverly construct dummy variables such that an estimated coefficient identifies this difference.

Using Dummy Variables for Multiple Categories

- What if we want to test the difference between middle and upper class?
- Cleverly construct dummy variables such that an estimated coefficient identifies this difference.

	Dummy Variables		
Social Class	$\tilde{D}_1$	$\tilde{D}_2$	
lower	0	0	$\hat{Y} = \hat{\beta}_0$
middle	1	0	$\hat{Y} = \hat{\beta}_0 + \hat{\beta}_1$
upper	1	1	$\hat{Y} = \hat{\beta}_0 + \hat{\beta}_1 + \hat{\beta}_2$

Using Dummy Variables for Multiple Categories

- What if we want to test the difference between middle and upper class?
- Cleverly construct dummy variables such that an estimated coefficient identifies this difference.

	Dummy Variables		
Social Class	$\tilde{D}_1$	$\tilde{D}_2$	
lower	0	0	$\hat{Y} = \hat{\beta}_0$
middle	1	0	$\hat{Y} = \hat{\beta}_0 + \hat{\beta}_1$
upper	1	1	$\hat{Y} = \hat{\beta}_0 + \hat{\beta}_1 + \hat{\beta}_2$

- When estimating $\hat{Y} = \hat{\beta}_0 + \hat{\beta}_1 \tilde{D}_1 + \hat{\beta}_2 \tilde{D}_2$ then the estimated coefficient of the second dummy, $\hat{\beta}_2$, represents (by design!) the difference between middle and upper class.

Interactions

Modeling Interactions

- So far, we have only been adding variables in an additive manner:

$$Y = \beta_0 + \beta_1 X_1 + \beta_2 X_2 + ... + \epsilon$$

- Suppose we want to test a hypothesis that the relation between an independent variable X_i and a dependent variable Y depends on the value of another dummy variable D.

Constituent term

- Think of: Income $= \beta_0 + \beta_1$Education $+ \beta_2$Female $+ \beta_3$Education $\times$ Female $+ \epsilon$
- The effect of X_1 on Y is also called conditional because the hypothesized effect is conditional on D.
- In other words, if D is 1, the relation between X_1 and Y is different than when D is zero.
- This is what we also call an interaction effect.

Modeling Interactions: Interpretation

- An interaction effect conditions the effect of an independent variable (e.g., Education) on the dependent variable (e.g., Income).
- Interaction model if D = 0 (condition is absent):

$$Y = \beta_0 + \beta_1 \cdot X_1 + \beta_2 \cdot 0 + \beta_3 \cdot X_1 \cdot 0 + \epsilon = \beta_0 + \beta_1 X_1 + \epsilon$$

- Interaction model if D = 1 (condition is present):

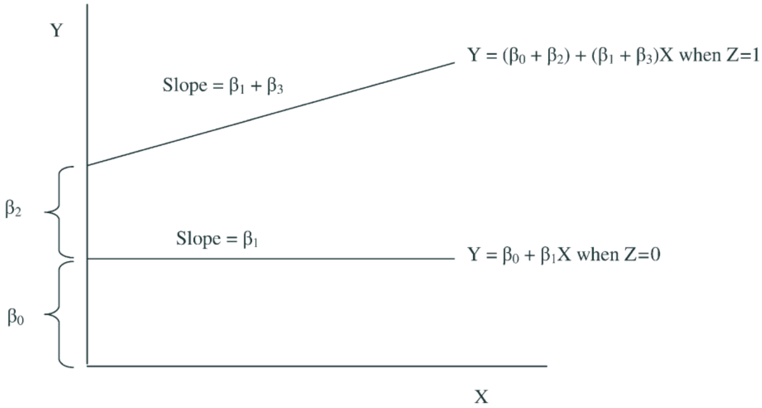
$$Y = \beta_0 + \beta_1 \cdot X_1 + \beta_2 \cdot 1 + \beta_3 \cdot X_1 \cdot 1 + \epsilon = (\beta_0 + \beta_2) + (\beta_1 + \beta_3)X_1 + \epsilon$$

- In other words, we get an intercept shift and a change in slopes.
- Do not interpret constitutive terms (i.e., $\hat{\beta}_1$ and $\hat{\beta}_2$ as if they are unconditional effects!)

Modeling Interactions: Interpretation

$$Y = \beta_0 + \beta_1 X + \beta_2 Z + \beta_3 XZ + \varepsilon$$

Hypothesis H_1: An increase in X is associated with an increase in Y when condition Z is met, but not when condition Z is absent.



Modeling Interactions: Interpretation

$$\hat{Y}_i = \beta_0 + \beta_1 X_i + \beta_2 M_i + \beta_3 (X_i M_i) + \epsilon_i$$

The interpretation of the regression coefficients changes:

B2 is the average change in income when gender goes from male to female, when education is 0

- β_0 is the expected value of Y when $X = 0$ and $M = 0$
- β_1 is the average change in Y when X increases by one unit, when M = 0.
- β_2 is the average change in Y when M increases by one unit, when $X = 0$.
- β_3 is the *interaction term* of X and M.

B3 is how much the impact of education on income changes when gender changes

If we rearrange the terms, we get:

$$\hat{Y}_i = \beta_0 + \beta_2 M_i + (\beta_1 + \beta_3 M_i) X_i + \epsilon_i$$

β_3 is the added increase in β_1, if M increases by one unit.

Modeling Interactions with Continuous Variables

- So far we studied interactions with dummy variables.
- But, we can interact continuous variables as well.
- Assume instead of a dummy D, X_2 to be continuous.
- Example: *Close (in time) presidential elections will reduce the effective number of electoral parties in subsequent parliamentary elections if and only if the number of presidential candidates is sufficiently low.*
- Thus, Number of parties regressed on the proximity of presidential

$$\text{Electoral Parties} = \beta_0 + \beta_1 \text{Proximity} + \beta_2 \text{Presidential Candidates} + \beta_3 \text{Proximity} \times \text{Presidential Candidates} + \epsilon$$

Modeling Interactions with Continuous Variables

- In this case, the effect of the explanatory variable on the outcome variable gradually changes as another variable changes.

$$Y = \beta_0 + \beta_1 X_1 + \beta_2 X_2 + \beta_3 X_1 X_2 + \ldots + \epsilon$$

$$Y = \beta_0 + \beta_2 X_2 + (\beta_1 + \beta_3 X_2) \cdot X_1 + \epsilon$$

- Marginal effect of X_1 on Y represents the effect of change in X_1 on the expected change in Y, especially when the change in the independent variable (X_1) is infinitely small (marginal).

Modeling Interactions with Continuous Variables

The marginal effect of proximity on the number of electoral parties is -2.66 for every presidential candidate added

Table 1 The impact of presidential elections on the effective number of electoral parties. Dependent variable: Effective number of electoral parties

Regressor	*Model*
Proximity	−3.44** (0.49)
PresidentialCandidates	0.29* (0.07)
Proximity*PresidentialCandidates	0.82** (0.22)
Controls	——
Constant	3.01** (0.33)
R^2	0.34
N	522

$^*p < 0.05$; $^{**}p < 0.01$ (two-tailed). Control variables not shown here. Robust standard errors clustered by country in parentheses.

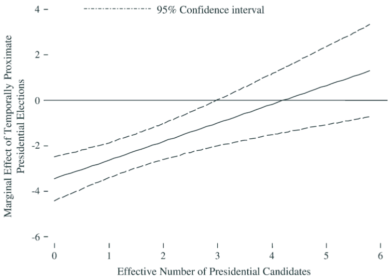


Fig. 3 The marginal effect of temporally proximate presidential elections on the effective number of electoral parties.

On average, a one unit increase in proximity results in a 3.44 unit decrease in number of electoral parties

The marginal effect of number of presidential candidates on the impact of proximity on the effective number of electoral parties (proximity + proximity*presidential candidates)

In-class example:

$$\text{Environmental Performance}_i = \alpha + \beta_1 \text{Income}_i + \beta_2 \text{Regime Type}_i + \beta_3 \text{Income}_i * \text{Regime Type}_i + \epsilon_i$$

```r
# Run regression model with interaction term
int_m2 <- lm(epi_epi ~ income + democracy + income*democracy, data = qog_data)

# Print results
summary(int_m2)
```

The marginal effect of a one unit increase in income on environmental performance is 7.29 for every democracy

this increases the effect of income on environmental performance by 5.10 from autocracies

```
## Coefficients:
##                          Estimate Std. Error t value Pr(>|t|)
## (Intercept)               37.1474     1.0684  34.768  < 2e-16 ***
## income                     2.1902     0.4532   4.833 3.24e-06 ***
## democracyDemocracy         3.4490     2.7819   1.240    0.217
## income:democracyDemocracy  5.1029     0.8686   5.875 2.55e-08 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 9.046 on 153 degrees of freedom
##   (37 observations deleted due to missingness)
## Multiple R-squared:  0.6879, Adjusted R-squared:  0.6818
## F-statistic: 112.4 on 3 and 153 DF,  p-value: < 2.2e-16
```

In-class example:

```
## Coefficients:
##                            Estimate Std. Error t value Pr(>|t|)
## (Intercept)                 37.1474     1.0684  34.768  < 2e-16 ***
## income                       2.1902     0.4532   4.833 3.24e-06 ***
## democracyDemocracy           3.4490     2.7819   1.240    0.217
## income:democracyDemocracy   5.1029     0.8686   5.875 2.55e-08 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 9.046 on 153 degrees of freedom
##   (37 observations deleted due to missingness)
## Multiple R-squared:  0.6879, Adjusted R-squared:  0.6818
## F-statistic: 112.4 on 3 and 153 DF,  p-value: < 2.2e-16
```

In-class example:

```
## Coefficients:
##                            Estimate Std. Error t value Pr(>|t|)    
## (Intercept)                 37.1474     1.0684  34.768  < 2e-16 ***
## income                       2.1902     0.4532   4.833 3.24e-06 ***
## democracyDemocracy           3.4490     2.7819   1.240    0.217    
## income:democracyDemocracy    5.1029     0.8686   5.875 2.55e-08 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 9.046 on 153 degrees of freedom
##   (37 observations deleted due to missingness)
## Multiple R-squared:  0.6879, Adjusted R-squared:  0.6818
## F-statistic: 112.4 on 3 and 153 DF,  p-value: < 2.2e-16
```

- The average Environmental Protection Index (EPI) for poor (Income = 0) autocracies is 37.1474 scale points (α).
- For autocracies, with every additional 10,000 USD of income, the EPI increases by 2.1902 scale points, on average (β_1) $\rightarrow$ Income effect for autocracies.
- For poor democracies, the EPI is 3.4490 scale points higher on average, in comparison to poor autocracies (β_2).
- For democracies, with every additional 10,000 USD of income, the EPI increases by 7.2931 scale points ($\beta_1 + \beta_3 = 2.1902 + 5.1029 = 7.2931$) $\rightarrow$ Income effect for democracies.

The systematic components:

Model for Autocracies (*democracy* = 0):

$$
\begin{aligned}
\hat{Y}_i &= 37.1474 + (2.1902 \cdot \text{Income}_i) + (3.4490 \cdot \text{Regime Type}_i) \\
&\quad + (5.1029 \cdot \text{Income}_i \cdot \text{Regime Type}_i) \\
\hat{Y}_i &= 37.1474 + (2.1902 \cdot \text{Income}_i) + (3.4490 \cdot 0) + (5.1029 \cdot \text{Income}_i \cdot 0) \\
\hat{Y}_i &= 37.1474 + 2.1902\,\text{Income}_i
\end{aligned}
$$

Model for Democracies (*democracy* = 1)

$$
\begin{aligned}
\hat{Y}_i &= 37.1474 + (2.1902 \cdot \text{Income}_i) + (3.4490 \cdot \text{Regime Type}_i) \\
&\quad + (5.1029 \cdot \text{Income}_i \cdot \text{Regime Type}_i) \\
\hat{Y}_i &= 37.1474 + (2.1902 \cdot \text{Income}_i) + (3.4490 \cdot 1) + (5.1029 \cdot \text{Income}_i \cdot 1) \\
\hat{Y}_i &= 40.5964 + (7.2931 \cdot \text{Income}_i)
\end{aligned}
$$

Visualization of example: